Introduction

Today we cover the material that appears in the end-of-semester paper, so write every formula down and check

1. Units, dimensions and measurement

Every physical quantity has a dimension and a unit; the SI unit of length is the metre, of mass the kilogram, and

Dimensional analysis checks an equation by comparing the dimensions on both sides: `[v] = L T^-1` and `[a] = L

A measurement of 25.4 mm is the same as 2.54 cm and 0.0254 m.

The uncertainty of a single reading with this ruler is 0.5 mm, so 25.4 mm is written as 25.4 ± 0.5 mm.
Significant figures follow the instrument: 3 significant figures for this ruler, not 6.
Worked examples:
- Convert 72 km/h to m/s: 72 × 1000 / 3600 = 20 m/s.
- A rod measures 1.25 m with a 1 mm tape, giving a percentage uncertainty of 0.1 / 125 = 0.08%.
Remember: this topic contributed 2 questions in last year's paper, and the examiner marked the unit errors strictl

2. Displacement, velocity and acceleration

Average velocity is displacement divided by time: `v = ”x / ”t`.

Acceleration is the rate of change of velocity: `a = ”v / ”t`, measured in m/s^2.

Velocity is a vector, so reversing direction changes its sign even when the speed is constant.

On a displacement-time graph the slope is velocity; on a velocity-time graph the slope is acceleration and the are

Uniform acceleration means the velocity changes by the same amount every second.

Worked examples:
- A bus covers 240 m in 12 s, so its average velocity is 20 m/s.
- A car speeds up from 8 m/s to 20 m/s in 4 s, so a = (20 - 8) / 4 = 3 m/s^2.
Remember: this topic contributed 3 questions in last year's paper, and the examiner marked the unit errors strictl
3. Equations of motion
For constant acceleration the four equations of motion are `v = u + at`, `s = ut + 1/2 at^2`, `v^2 = u^2 + 2as` and

u is the initial velocity, v the final velocity, a the acceleration, t the time and s the displacement.
Each equation leaves out exactly one quantity, so choose the equation that omits what you are not given.
The equations only hold while the acceleration stays constant.
For free fall near the Earth's surface a = 9.8 m/s^2 downward, often rounded to 10 m/s^2 in estimates.
Worked examples:
- A stone dropped from rest falls 45 m: 45 = 1/2 × 10 × t^2, so t = 3 s and v = 30 m/s.

- A train at 15 m/s brakes at 2 m/s^2 and stops after 56.25 m, because v^2 = u^2 + 2as gives 0 = 225 - 4s.
Remember: this topic contributed 4 questions in last year's paper, and the examiner marked the unit errors strictl
4. Newton's laws of motion
Newton's first law: a body stays at rest or in uniform motion unless a resultant force acts on it.
Newton's second law: the resultant force equals mass times acceleration, `F = ma`, with F in newtons.
Newton's third law: every action has an equal and opposite reaction, acting on a different body.

One newton accelerates 1 kg at 1 m/s^2.
Weight is a force, `W = mg`, so a 60 kg student weighs 588 N on Earth.
Worked examples:
- A 1200 kg car accelerates at 2 m/s^2, so the resultant force is 2400 N.
- A 5 kg block on a frictionless table pulled by 15 N accelerates at 3 m/s^2.
Remember: this topic contributed 5 questions in last year's paper, and the examiner marked the unit errors strictl

5. Friction and inclined planes

Static friction rises to a maximum of `f = ¼N` before the surfaces slide.

Kinetic friction is usually smaller than the maximum static friction.

The normal reaction N is perpendicular to the surface, not always equal to the weight.

On a slope of angle ¸ the component of weight along the slope is `mg sin ¸` and the normal reaction is `mg cos ¸`

The coefficient of friction ¼ has no unit.

Worked examples:

- A 10 kg crate on a floor with ¼ = 0.4 needs 39.2 N to start moving, since f = 0.4 × 98 N.
- On a 30 degree slope with no friction a block accelerates at 4.9 m/s^2, from 9.8 × sin 30.

Remember: this topic contributed 6 questions in last year's paper, and the examiner marked the unit errors strictl

6. Work, energy and power

Work is force times displacement along the force: `W = Fs cos ¸`, measured in joules.

Kinetic energy is `KE = 1/2 mv^2` and gravitational potential energy is `PE = mgh`.
The work-energy theorem says the resultant work equals the change in kinetic energy.
Power is the rate of doing work: `P = W / t`, measured in watts.
Energy is conserved in a closed system, though it may change form.
Worked examples:
- A 2 kg ball at 6 m/s has KE = 1/2 × 2 × 36 = 36 J.

- Lifting 20 kg by 3 m takes 588 J of work, and doing it in 2 s needs 294 W.

Remember: this topic contributed 7 questions in last year's paper, and the examiner marked the unit errors strictl

7. Momentum and collisions

Momentum is `p = mv`, measured in kg m/s and pointing along the velocity.

In a collision with no external force, total momentum before equals total momentum after.

Impulse is `F”t = ”p`, the change in momentum.

An elastic collision conserves kinetic energy; an inelastic one does not.
A perfectly inelastic collision joins the bodies, so they share one final velocity.
Worked examples:
- A 0.15 kg ball at 20 m/s has momentum 3 kg m/s.
- A 2 kg trolley at 3 m/s hits a stationary 4 kg trolley and they stick: v = 6 / 6 = 1 m/s.
Remember: this topic contributed 8 questions in last year's paper, and the examiner marked the unit errors strictl

8. Circular motion

Angular velocity is `É = 2À / T`, measured in rad/s.

Centripetal acceleration is `a = v^2 / r = É^2 r`, always pointing to the centre.

The centripetal force is `F = mv^2 / r`, supplied by tension, friction or gravity.

Velocity is tangential while acceleration is radial, so the speed can stay constant.

One full revolution is 2À radians, or 360 degrees.

Worked examples:
- A 0.5 kg mass on a 2 m string at 4 m/s needs 4 N inward, from 0.5 × 16 / 2.
- A wheel with T = 0.5 s has É = 4À rad/s, about 12.6 rad/s.
Remember: this topic contributed 9 questions in last year's paper, and the examiner marked the unit errors strictl
9. Simple harmonic motion
In simple harmonic motion the acceleration is proportional to the displacement and opposite in direction: `a = -É^

The period of a spring is `T = 2À  (m/k)` and of a simple pendulum `T = 2À  (L/g)`.
Amplitude is the maximum displacement; the period does not depend on amplitude.
At maximum displacement the velocity is zero and the acceleration is greatest.
At the equilibrium position the velocity is greatest and the acceleration is zero.
Worked examples:
- A pendulum of length 1 m has T = 2À (1 / 9.8) = 2.0 s.

- A 0.2 kg mass on a spring with k = 20 N/m oscillates with T = 0.63 s.

Remember: this topic contributed 10 questions in last year's paper, and the examiner marked the unit errors stric

10. Waves and sound

The wave equation is `v = f»`, so a 440 Hz note with » = 0.78 m travels at 343 m/s.

Frequency is the number of oscillations per second, measured in hertz.

Sound needs a medium, and its speed is about 343 m/s in air at 20 degrees Celsius.

Diffraction is most noticeable when the gap is about the same size as the wavelength.
Resonance occurs when a driving frequency matches the natural frequency.
Worked examples:
- A wave of period 0.02 s has f = 50 Hz, and at 10 m/s its wavelength is 0.2 m.
- A pipe closed at one end resonates at odd multiples: 100 Hz, 300 Hz, 500 Hz.
Remember: this topic contributed 11 questions in last year's paper, and the examiner marked the unit errors stric

11. Electricity: current, voltage and resistance

Current is the rate of flow of charge: `I = Q / t`, measured in amperes.

Ohm's law states `V = IR` for a conductor at constant temperature.

Resistors in series add directly; in parallel the reciprocals add: `1/R = 1/R1 + 1/R2`.

Electrical power is `P = VI = I^2 R = V^2 / R`, measured in watts.

One coulomb is the charge carried by 1 A in 1 s.

Worked examples:
- A 12 V supply across 4 © drives 3 A, using 36 W.
- Two 6 © resistors in parallel give 3 ©.
Remember: this topic contributed 12 questions in last year's paper, and the examiner marked the unit errors stric
12. Heat and thermodynamic energy
Heat capacity is `Q = mc”T`, with c in J/(kg K).

The specific heat capacity of water is 4180 J/(kg K).
Latent heat changes state without changing temperature: `Q = mL`.
The specific latent heat of fusion of ice is about 334 000 J/kg.
Temperature measures average kinetic energy, while internal energy is the total.
Worked examples:
- Heating 2 kg of water by 10 K needs 83 600 J.

- Melting 0.5 kg of ice needs 167 000 J.

Remember: this topic contributed 13 questions in last year's paper, and the examiner marked the unit errors stric

13. Pressure, density and fluids

Pressure is force per unit area: `p = F / A`, measured in pascals.

Pressure in a liquid increases with depth: `p = Ágh`.

Density is mass per unit volume: `Á = m / V`.

Archimedes' principle: the upthrust equals the weight of fluid displaced.
A body floats when its average density is less than the fluid's density.
Worked examples:
- At 5 m depth in water the gauge pressure is 1000 × 9.8 × 5 = 49 000 Pa.
- A 2 kg block of volume 0.001 m^3 has density 2000 kg/m^3 and sinks in water.
Remember: this topic contributed 14 questions in last year's paper, and the examiner marked the unit errors stric

14. Gases and the kinetic model

The ideal gas equation is `PV = nRT`, with R = 8.314 J/(mol K).

For a fixed mass at constant temperature, `P1V1 = P2V2` (Boyle's law).

At constant pressure, `V1/T1 = V2/T2` (Charles's law).

Temperature in the gas laws must be in kelvin, where 0 degrees Celsius is 273 K.

The kinetic model says pressure comes from molecules colliding with the walls.

Worked examples:
- 2 mol of gas at 300 K in 0.05 m^3 exerts 99 768 Pa.
- Doubling the pressure of 3 L of gas at constant temperature gives 1.5 L.
Remember: this topic contributed 15 questions in last year's paper, and the examiner marked the unit errors stric
Revision plan
Work through chapters 1 to 9 of the handbook, then attempt the past paper under timed conditions. Bring your fo

That is all for today. Read pages 112 to 148 before the next lecture.